Final assessment 2026 Economics and Society

Part 1 [15 marks]

1.11 AI prompt

Generate a 300-word summary of the UK government proposal to apply VAT to private school fees. Explain the main policy change and discuss its potential economic costs and benefits.

1.12 Chat GPT output

The UK government has proposed a policy to apply Value Added Tax (VAT) to private school fees. This proposal would remove the existing VAT exemption for private education and require private schools to charge VAT at the standard rate on tuition fees. The primary purpose of the policy is to generate additional government revenue and to eliminate disparities in tax treatment between private and state education. The government has stated that the revenue raised could be used to increase spending on state schools.

The policy has several potential economic benefits. One benefit is that applying VAT to private school fees would broaden the tax base. This may improve the consistency of the tax system, as most goods and services are already subject to VAT. Another benefit is that the policy could raise a significant amount of revenue for the government. This revenue could be used to fund public services, particularly education. In addition, higher private school fees may reduce demand for private education. This could lead to a shift of some pupils into the state school sector, potentially increasing social integration.

However, there are also potential economic costs associated with the policy. Demand for private schooling may be inelastic, meaning that many families may continue to pay private school fees despite higher prices. As a result, the tax burden may fall on parents. Private schools may experience lower enrolment, which could lead to financial difficulties, school closures, or job losses. If some pupils move from private to state schools, this could increase pressure on state school capacity and public spending. There may also be administrative costs related to implementing VAT for private schools.

Overall, the proposal to apply VAT to private school fees has both potential benefits and potential costs. The policy involves trade-offs between revenue generation, efficiency, equity, and impacts on the education system.

1.2 Comments on accuracy

The AI-generated output provides a reasonable overview of the proposed VAT reform, but while many of its claims are directionally correct, several are either incomplete or overstated when assessed against the IFS report and course material.

The description of the policy itself is accurate. The AI correctly states that private school fees are currently exempt from VAT and that the reform would remove this exemption. This is directly supported by the IFS, which states that “independent school fees are currently VAT exempt” and that Labour proposes levying VAT on these fees (IFS, Executive Summary, p.3). As a factual description of the policy change, the output is therefore dependable.

However, the AI output becomes less accurate when discussing the size of the price increase faced by households. By referring to VAT at the standard rate without further qualification, it implicitly suggests a 20% rise in fees. The IFS explicitly cautions against this interpretation, noting that “the effective VAT rate would be around 15% rather than the full 20%” once input deductions and exemptions are considered (IFS, Executive Summary, p.3; Table 2, pp.16-17). This distinction matters economically, as overstating the price increase can lead to exaggerated predictions about demand responses.

The claim that the policy would raise substantial government revenue is supported by the evidence. The IFS estimates gross revenues of around £1.6 billion per year, with a net gain of £1.3-£1.5 billion after allowing for additional state school spending (IFS, pp.24-25). From a public finance perspective, this suggests the revenue argument for the policy is strong and credible. (Mirrlees et al., 2011).

Where the AI output is weakest is in its treatment of behavioural effects. It suggests that higher fees could trigger widespread school closures and major disruption. The IFS evidence does not support this. Based on observed behaviour and elasticity estimates, the report suggests a 3-7% fall in private school attendance, equivalent to 20,000–40,000 pupils, occurring gradually over time (IFS, pp.21-22). The IFS also notes that demand for private schooling appears price-inelastic, implying that many families will absorb higher costs rather than exit the sector, making “large-scale closures unlikely in the short run” (IFS, pp.21-23).

In the course's terms, the policy debate is best framed as an equity–efficiency trade-off: removing a VAT exemption targets a service disproportionately consumed by higher-income households, while raising funds for a publicly provided service with large distributional implications. This links directly to the course discussion of why public and private schooling co-exist and how a two-tier system can amplify inequality even when ‘choice’ is given.

Overall, while the AI output captures the broad trade-offs involved in the policy, it lacks the nuance provided by empirical evidence. Applying concepts from the course, particularly tax incidence and elasticity of demand, suggests that the policy is more likely to raise revenue with limited behavioural disruption than the AI output implies. (Atkinson and Stiglitz, 1980; Mirrlees et al., 2011).

1.3 Rewritten summary

The UK government has proposed applying Value Added Tax (VAT) to private school fees by removing the existing VAT exemption for private education. While the statutory VAT rate is 20%, evidence from the Institute for Fiscal Studies (IFS) suggests that the effective increase in fees would be closer to 15%, once deductions for input VAT, exemptions for specialist provision, and the treatment of boarding fees are taken into account (IFS, Executive Summary, p.3; Table 2, pp.16-17). From a public finance perspective, the policy is primarily motivated by revenue generation and by concerns about horizontal and vertical equity in the tax system, rather than by an explicit attempt to eliminate private schooling.

A key economic benefit of the policy is revenue generation. According to the IFS estimates, removing tax exemptions from private schools would raise around £1.6 billion per year, with a net gain of £1.3-£1.5 billion after accounting for additional state school spending (IFS, pp.24-25). Since private school attendance is heavily concentrated among higher-income households, applying VAT to private school fees is likely to increase the progressivity of the tax system, aligning with equity-based arguments discussed in the course. (IFS, 2024)

Concerns have been raised that higher fees could significantly reduce demand for private schooling. However, the evidence presented in the IFS report suggests that demand for private education is price inelastic. Based on plausible elasticity estimates, the policy is expected to result in a 3-7% reduction in private school enrolment, equivalent to 20,000–40,000 pupils, occurring gradually over time rather than immediately (IFS, pp.21-22). As a result, large-scale private school closures are considered unlikely in the short run (IFS, pp.21-23).

If some pupils move into the state sector, this will create additional public spending pressures. However, the IFS estimates that the required increase in state school spending would be modest relative to the revenue raised, suggesting that the policy would still result in a substantial net fiscal gain (IFS, pp.24-25).

Overall, once realistic behavioural responses and effective tax rates are considered, the policy appears far more likely to function as a revenue-raising and redistributive instrument than as a disruptive reform of the education system. The VAT reform is likely to raise significant revenue, improve tax fairness, and generate limited disruption to the education system. From an economic perspective, this suggests that much of the burden of the tax is likely to be borne by higher-income households rather than translated into large behavioural responses.

Part 2 [40 marks]

2.1 [5 marks]

Let private school tuition fees be t, The VAT rate is τ. Let y be the number of families attending private school and F the total fee revenue received by the private school. Since each private-school family pays tuition t, total fee revenue is.

$$F = yt.$$

For a family of type $i \in \{L, M, H\}$, the gross benefit from private school is given by:

$$U_i^{benefit} = P_i + 10\sqrt{\frac{F}{y}}.$$

Because,

$$\frac{F}{y} = \frac{yt}{y} = t,$$

This simplifies (regardless of how many families attend) to:

$$u_i = P_i + 10\sqrt{t}.$$

Note that because $F/y = t$, the private-school 'quality' term does not depend on the number of private-school families y. This implies no congestion or peer effect in private schooling within this model; all strategic interaction operates through the public-school crowding term. S/x and the redistribution term $5T/x$.

The monetary cost of private schooling is tuition plus VAT:

$$Cost = t + \tau t = t(1 + \tau).$$

Therefore, the payoff from sending the child to private school is:

$$U_i^{priv} = u_i - t(1 + \tau) = P_i + 10\sqrt{t} - t(1 + \tau),$$

$$i \in \{L, M, H\}.$$

Explicitly:

$$U_L^{priv} = P_L + 10\sqrt{t} - t(1 + \tau),$$

$$U_M^{priv} = P_M + 10\sqrt{t} - t(1 + \tau), U_H^{priv} = P_H + 10\sqrt{t} - t(1 + \tau).$$

Economic interpretation of each term

$P_L < P_M < P_H$ are type-specific valuations of private schooling, capturing that higher-income families obtain larger private benefits (e.g. preferences, ability to pay, perceived returns). $10\sqrt{t}\ It$ is a quality term that increases with resources but exhibits diminishing returns. Finally, $t(1 + \tau)$ is the VAT-inclusive price faced by families; increasing τ raises the private cost directly, discouraging private-school attendance through the price channel, while any quality response operates separately through VAT-financed public-school funding. The payoff structure highlights that VAT operates purely through prices on the private side, while quality responses arise only through public funding.

This structure mirrors the general choice framework introduced in lecture four, where agents compare net utilities across discrete alternatives. Here, VAT affects behaviour exclusively through the price channel on the private side, while distribution operates through publicly provided quality, consistent with the course's treatment of policy-induced price distortions. (Mirrlees et al., 2011). This is shown in the standard models of consumption taxation, where VAT enters as a wedge between marginal benefit and marginal cost.

The private-school payoff consists of a type-based valuation. P_i, a diminishing-returns quality component $10\sqrt{t}$, and the VAT-inclusive price $t(1 + \tau)$. A higher VAT rate reduces U_i^{priv} and therefore weakens incentives to choose private schooling.

2.2 [7.5 marks]

In scenario one, the low-income family L attends public school while the middle and high-income families M and H attend private. Hence,

$$x = 1 \; and \; y = 2.$$

This equilibrium concept follows the static Nash equilibrium framework discussed in Lecture 5, where agents choose simultaneously and no individual has an incentive to deviate given others' choices.

VAT revenue and public-school payoff

Total VAT revenue equals VAT per private pupil (τt) times the number of private pupils (y):

$$T = \tau t y = 2\tau t.$$

Public-school utility (for any family in public school) is given by:

$$U^{pub} = \frac{S}{x} + \frac{5T}{x}.$$

Substituting $x = 1$ and $T = 2\tau t$,

$$U^{pub}(1,2) = S + 10\tau t.$$

Hence L's payoff in public under Scenario 1 is:

$$U_L^{pub} = S + 10\tau t.$$

Nash equilibrium conditions

To characterise equilibrium, it is sufficient to check unilateral deviation, since families choose schools simultaneously and take others' choices as given.

i) Low-income family

If L deviates to private school, then all three families attend private school, $x = 0$ and $y = 3$, so public school utility is not defined, and L must compare her private payoff to her current public payoff under scenario one. The private payoff for L would still be:

$$U_L^{priv} = P_L + 10\sqrt{t} - t(1 + \tau).$$

Therefore L stays in public if:

$$U_L^{pub}(1,2) \geq U_L^{priv} \Leftrightarrow S + 10\tau t \geq P_L + 10\sqrt{t} - t(1 + \tau). \qquad (\mathrm{S1 - L})$$

ii) Middle-income family

If M deviates to public school, then $x = 2$, $y = 1$, and VAT revenue becomes,

$$T' = \tau t \cdot 1 = \tau t.$$

So, the public payoff upon deviation is:

$$U^{pub}(2,1) = \frac{S}{2} + \frac{5T'}{2} = \frac{S}{2} + \frac{5\tau t}{2}.$$

Thus M remains in private if:

$$U_M^{priv} \geq U^{pub}(2,1) \Leftrightarrow P_M + 10\sqrt{t} - t(1+\tau) \geq \frac{S}{2} + \frac{5\tau t}{2}. \qquad \text{(S1-M)}$$

iii) High-income family

The same deviation environment applies, so Hstays private if:

$$U_H^{priv} \geq U^{pub}(2,1) \Leftrightarrow P_H + 10\sqrt{t} - t(1+\tau) \geq \frac{S}{2} + \frac{5\tau t}{2}. \qquad \text{(S1-H)}$$

Since $P_H > P_M$, condition (S1-H) is weaker than (S1-M). Hence the binding "upper bound" is driven by M's incentive.

Parameter range

Rearranging (S1-L) gives a lower bound on S:

$$S \geq P_L + 10\sqrt{t} - t(1+\tau) - 10\tau t.$$

Rearranging (S1-M) gives an upper bound on S:

$$S \leq 2\left(P_M + 10\sqrt{t} - t(1+\tau)\right) - 5\tau t.$$

Therefore, Scenario 1 is a Nash equilibrium whenever,

$$P_L + 10\sqrt{t} - t(1+\tau) - 10\tau t \;\leq\; S \;\leq\; 2\left(P_M + 10\sqrt{t} - t(1+\tau)\right) - 5\tau t,$$

With,

$$0 \leq P_L < P_M < P_H.$$

Numerical example

For simplicity, units are interpreted so that tuition is to be measured in tens rather than pounds.

Let:

$$t = 10, \qquad \tau = 0.2, \qquad P_L = 2, \qquad P_M = 5, \qquad P_H = 8\,, \qquad S = 20.$$

Then:

$$10\sqrt{t} \approx 31.623, \qquad t(1+\tau) = 12.$$

Private utilities:

$$U_L^{\text{priv}} \approx 21.62, \qquad U_M^{\text{priv}} \approx 24.62, \qquad U_H^{\text{priv}} \approx 27.62.$$

Public utilities:

$$U_L^{\text{pub}} = 20 + 20 = 40, U^{\text{pub}}(2,1) = 10 + 5 = 15.$$

Thus L prefers public school, while M and H prefer private school. No family wishes to deviate, confirming Scenario 1 as a Nash equilibrium.

2.3 [7.5 marks]

Allocation

In scenario two, families L and M attend public school, while H attends private school.

Hence,

$$x = 2 \; and \; y = 1.$$

Public school payoff

VAT revenue is:

$$T = \tau t y = \tau t.$$

Public payoff utility is therefore:

$$U^{pub}(2,1) = \frac{S}{2} + \frac{5T}{2} = \frac{S}{2} + \frac{5\tau t}{2}.$$

Private-school payoff remains:

$$U_L^{priv} = P_L + 10\sqrt{t} - t(1+\tau),$$

$$i \in \{L, M, H\}.$$

Nash equilibrium conditions

i) Low-income family

$$U^{pub}(2,1) \geq U_L^{priv} \Leftrightarrow \frac{S}{2} + \frac{5\tau t}{2} \geq P_L + 10\sqrt{t} - t(1+\tau). \qquad \text{(S2-L)}$$

ii) Middle-income family (binding condition because $P_M > P_L$)

$$\frac{S}{2} + \frac{5\tau t}{2} \geq P_M + 10\sqrt{t} - t(1+\tau). \qquad \text{(S2-M)}$$

iii) High-income family

If H deviates to public, then $x = 3$, $y = 0$, so $T = 0$ and

$$U^{pub}(3,0) = \frac{S}{3}.$$

Thus Hremains private if:

$$U_H^{priv} \geq \frac{S}{3} \Leftrightarrow P_H + 10\sqrt{t} - t(1+\tau) \geq \frac{S}{3}. \qquad \text{(S2-H)}$$

Parameter range

Scenario two is a Nash equilibrium whenever:

$$2\left(P_M + 10\sqrt{t} - t(1+\tau)\right) - 5\tau t \;\leq\; S \;\leq\; 3\left(P_H + 10\sqrt{t} - t(1+\tau)\right).$$

Compared to scenario one, the lower bound on S is higher, reflection the fact that sustaining two families in public schooling requires stronger public provision.

Numerical example

Let:

$$t = 10, \qquad \tau = 0.2, \qquad P_L = 2, \qquad P_M = 5, \qquad P_H = 8, \qquad S = 42.$$

Public utilities:

$$U_{L,M}^{\text{pub}} = 21 + 5 = 26.$$

Private utilities unchanged:

$$U_L^{\text{priv}} \approx 21.62, \qquad U_M^{\text{priv}} \approx 24.62, \qquad U_H^{\text{priv}} \approx 27.62.$$

Deviation payoff for H:

$$U_H^{\text{pub}}(3,0) = 14.$$

Hence $L$and Mprefer public school and H prefers private school, confirming Scenario 2 as a Nash equilibrium.

2.4 [5 marks]

The comparative statics are most naturally understood by focusing on the middle-income family, whose decision differs across the two scenarios. The difference between Scenario 1 and Scenario 2 lies in the schooling decision of the middle-income family M. In Scenario 1, M attends private school, while in Scenario 2, M attends public school. This follows the approach outlined in Lecture 6, which emphasises analysing policy changes through marginal responses of agents at the boundary between choices, rather than focusing on infra-marginal behaviour. Consequently, this on how changes in the model's parameters affect the relative attractiveness of public versus private

schooling for family M, holding the behaviour of L and H fixed. (Epple and Romano, 1998; Nechyba, 2000).

Baseline public-school quality S

Public-school utility increases in S through the term S/x. An increase in S raises the payoff from public schooling for all families and makes the public-school participation constraints in Scenario 2 easier to satisfy. At the same time, a higher S makes it more difficult to sustain Scenario 1, since the middle-income family's payoff from deviating to public school increases. Therefore, higher S shifts the equilibrium towards Scenario 2, while lower S makes Scenario 1 more likely.

VAT rate τ

An increase in τ raises the private-school cost $t(1+\tau)$, reducing private-school utility, and simultaneously increases VAT revenue, which improves public-school quality through the term $5\tau ty/x$. Both effects reduce the relative attractiveness of private schooling for the middle-income family. Consequently, a higher VAT rate makes Scenario 2 more likely, whereas a lower VAT rate favours Scenario 1.

Tuition fees t

Changes in t affect both private-school quality, through the term $10\sqrt{t}$, and private-school costs, through $t(1+\tau)$, as well as public-school quality via VAT revenue. The effect of t on the equilibrium is therefore ambiguous. If the quality effect dominates, higher t can increase the attractiveness of private schooling and support Scenario 1. If the cost and tax effects dominate, higher t pushes the middle-income family towards public schooling and favours Scenario 2.

Type-specific valuations P_L, P_M, P_H

The parameter P_M plays a significant role since the middle-income family switches sectors across scenarios. A higher P_M raises the private-school payoff for M, making Scenario 1 more likely. A lower P_M makes public schooling more attractive, favouring Scenario 2. Changes in P_L and P_H mainly affect whether the low-income family continues to prefer public school and the high-income family continues to prefer private school; if these conditions hold, they do not determine which scenario arises.

Taken together, higher baseline public quality S and a higher VAT rate τ shift the equilibrium towards Scenario 2 (both L and M in public school), while lower S and lower τ favour Scenario 1 (only L in public school). The valuation parameter P_M tilts the balance between the two scenarios, and the effect of tuition fees t is ambiguous, depending on whether quality or cost effects dominate. Intuitively, this reflects that the middle-income family is the only group whose schooling decision responds on the extensive margin. Overall, the model predicts that policies improving public provision or raising private- school costs tend to shift the equilibrium along the extensive margin of middle-income participation. This highlights that the extensive margin of school choice is driven primarily by middle-income households rather than by uniform responses across all income groups.

2.5 [5 marks]

Let social welfare be defined as the sum of utilities of all three families:

$$W = U_L + U_M + U_H.$$

Effect of VAT on individual utilities

For a family attending private school, the payoff is:

$$U_i^{\text{priv}} = P_i + 10\sqrt{t} - t(1+\tau), i \in \{L, M, H\}.$$

Differentiating with respect to the VAT rate τ yields:

$$\frac{\partial U_i^{\text{priv}}}{\partial \tau} = -t < 0.$$

Thus, an increase in VAT reduces the utility of any family attending private school, since it raises the effective price of private education.

For a family attending public school, the payoff is:

$$U_i^{\text{pub}} = \frac{S}{x} + \frac{5\tau t y}{x}.$$

Differentiating with respect to τ,

$$\frac{\partial U_i^{\text{pub}}}{\partial \tau} = \frac{5ty}{x} > 0.$$

Hence, a higher VAT rate increases the utility of public-school families through increased public-school funding.

Effect on social welfare

Let x denote the number of families attending public school and $y = 3 - x$ the number attending private school. The total effect of a marginal increase in VAT on social welfare is:

$$\frac{dW}{d\tau} = x\left(\frac{5ty}{x}\right) + y(-t) = 5ty - ty = 4ty.$$

It is worth emphasising that this welfare result is local and holds only when the schooling allocation is fixed. It reflects the standard partial equilibrium welfare analysis from Lecture 7, where marginal tax changes are welfare improving when they relax a binding public finance constraint without inducing large behavioural distortions. (Mirrlees et al., 2011). This highlights the standard result that small tax changes can be welfare improving even when substantial changes are distortionary.

In conclusion, if at least one family attends a private school ($y > 0$), then $\frac{dW}{d\tau} > 0$, so a marginal increase in VAT raises social welfare. If no family attends private school ($y = 0$VAT has no effect on welfare since no revenue is raised. Welfare may decrease only if a sufficiently large VAT increase alters schooling choices and leads to congestion effects in public schools. Thus, small increases in VAT are welfare-enhancing, while large increases may not be. This result abstracts from general equilibrium effects such as changes in class size or teacher supply, which could matter for larger policy changes.

2.6 [10 marks]

Whether introducing VAT on private school fees is a desirable policy depends on the trade-off between equity and efficiency. (Atkinson and Stiglitz, 1980; Mirrlees et al., 2011). The theoretical model highlights two opposing effects of VAT. First, VAT increases the cost of private schooling to $t(1+\tau)$, reducing the utility of families who attend private schools. This effect is strongest for high- and middle-income families, who place higher valuations on private education ($P_H > P_M > P_L$). Second, VAT revenue is redistributed to the public education system, raising public-school quality via the term. $5\tau ty/x$. This benefits families who attend public schools, particularly low-income households.

In Scenario 1, where only the low-income family attends public school, a marginal increase in VAT raises social welfare if some families remain in private school. The result $\frac{dW}{d\tau} = 4ty > 0$ shows that the gains from improved public-school funding outweigh the losses borne by private-school families. This mirrors the distinction between efficiency and redistribution discussed in the lectures on taxation and public spending.

In Scenario 2, where both low- and middle-income families attend public school, the same mechanism applies, but the welfare gain from increasing VAT is smaller because fewer families pay the tax. Nonetheless, the policy continues to shift resources from higher-income households toward public education.

A potential drawback arises if VAT is set too high. Large increases in VAT may induce middle-income families to switch from private to public schooling, increasing congestion and reducing per-pupil resources in public schools. In this case, the efficiency costs of overcrowding may offset the redistributive benefits. This equity efficiency trade-off shows the broader policy evaluation framework, which was mentioned in Lecture 3, where redistributive gains must be weighed against potential efficiency losses arising from congestion and capacity restraints.

Overall, the model suggests that VAT on private schools can improve both equity and social welfare if implemented moderately and if the additional revenue is used effectively. However, excessively high VAT rates risk undermining efficiency by overburdening the public education system. The model supports VAT on private schooling as a potentially effective redistributive tool, but only when applied cautiously and alongside sufficient public-sector capacity.

Part 3 [30 marks]

3.1 [7.5 marks]

Below are testable predictions implied by the model in Part 2. Each prediction is stated in the required "when X increases, Y changes" format and links directly to the mechanisms in the payoff functions.

Prediction 1: VAT and private-school attendance

Holding other parameters fixed, an increase in the VAT rate τlowers the payoff from private schooling for any family:

$$\frac{\partial U_i^{priv}}{\partial \tau} = -t < 0,$$

and raises the payoff from public schooling for those in public school (when $y > 0$):

$$\frac{\partial U^{pub}}{\partial \tau} = \frac{5ty}{x} > 0.$$

Therefore, when τ increases, private-school attendance (the share y) decreases, especially among "marginal" families (most plausibly middle-income families).

Prediction 2: Public-school baseline quality and school choice

An increase in the baseline public-school quality parameter S increases public-school utility through S/x.
Therefore, when S increases, public-school attendance (the share x) increases and private-school attendance falls, again most strongly among families close to indifference between sectors.

Prediction 3: Private-school fees and private-school attendance

Private-school utility contains a quality term $10\sqrt{t}$ (increasing in $t$with diminishing returns) and a cost term $-t(1+\tau)$(linear and increasing in t). Hence:

$$\frac{\partial U_i^{priv}}{\partial t} = \frac{5}{\sqrt{t}} - (1+\tau),$$

So, the sign is ambiguous and depends on parameter values.

Therefore:

If $(1+\tau) > \frac{5}{\sqrt{t}}$ (i.e. cost dominates), then higher t reduces private-school attendance.

If $(1+\tau) < \frac{5}{\sqrt{t}}$ (quality dominates), then higher t could increase private-school attendance.

This yields a conditional ("only if…") prediction, which reflects the competing quality and cost mechanisms highlighted by the model.

Prediction 4: Heterogeneity by income type

Since $P_H > P_M > P_L$Higher-income families have systematically higher private-school valuation. Therefore, when τ increases, high-income families are more likely to remain in private schooling than middle- and low-income families; switching from private to public is more concentrated among lower-valuation groups. This implies heterogeneous treatment effects of VAT by income group.

Prediction 5: VAT and social welfare (local prediction)

From Part 2, holding the allocation fixed:

$$\frac{dW}{d\tau} = 4ty.$$

Therefore, when τ increases, social welfare increases locally whenever $y > 0$ (i.e. when some families remain in private school). However, this prediction is local: sufficiently large VAT changes may alter the equilibrium allocation and potentially introduce congestion effects in public schools.

The key implication is that the model predicts that higher VAT rates and higher baseline public quality reduce private-school attendance, with stronger responses among marginal (middle-income) families, and it yields a conditional prediction that the effect of tuition t on private attendance depends on whether cost or quality effects dominate. These predictions are directly testable and map cleanly from the models to observable enrolment and price outcomes, following the positive analysis approach emphasised in Lecture 8, where theory is used to generate falsifiable empirical implications rather than normative conclusions. (Angrist and Pischke, 2009).

3.2 [7.5 marks]

This section maps theoretical variables into measurable real-world counterparts and identifies credible data sources to operationalise them.

School-choice outcomes (x and y). The model's central outcomes are the numbers (or shares) in public versus private schooling. Empirically, y can be measured as the share (or count) of pupils in independent schools, and x as the complementary share in state-funded schools. (Institute for Fiscal Studies, 2024; OECD, 2019). Independent-school pupil counts and regional breakdowns can be obtained from the Independent Schools Council (ISC) annual census reports, which provide pupil totals and changes over time. If y is the share of pupils in independent schools, this can be measured using the Independent Schools Council census, which covers over 1,400 schools and reports detailed pupil numbers and characteristics. (Independent Schools Council, 2025). Public-school enrolment, class sizes, and pupil characteristics are available from the Department for Education (DfE) School Census statistical releases. (Department for Education, 2025).

VAT policy variable (τ). The policy can be coded as a post-reform indicator (0 before VAT applies to fees; one after), and/or as the statutory VAT rate applied to fees. (Institute for Fiscal Studies, 2024). The mechanism is that τ raises the VAT-inclusive price $t(1 + \tau)$. (Mirrlees et al., 2011).

Private-school fees (t). The theoretical tuition fee t can be proxied by average private-school fees (day and boarding) reported in ISC publications (or fee schedules) and then converted into a VAT-inclusive measure by multiplying by $(1 + \tau)$. (Independent Schools Council, 2025). The relevant regressor in price-based specifications is thus $t(1 + \tau)$. (Institute for Fiscal Studies, 2024).

Public-school "quality/resources" (S and VAT-funded spending). The model captures public quality with S and with VAT-financed funding per pupil. In practice, these can be substituted using class size measures and other pupil/resource statistics from DfE releases. (OECD, 2019; Glennerster, 2003). When focusing on outcomes rather than inputs, public-school performance can be proxied using inspection and performance measures.

Income heterogeneity (proxying P_L, P_M, P_H). Since the P_i terms are unobserved valuations that vary with income; income type can be represented using local-area household income. (OECD, 2019; Glennerster, 2003). A suitable source is the ONS small area income estimates, which provide household income measures at the MSOA/local area level and can be matched geographically to education outcomes.

Outcome proxies related to welfare (optional extension). Social welfare is not directly observed, but model-relevant welfare proxies include public-school outcomes (e.g., inspection outcomes). (Atkinson and Stiglitz, 1980). Ofsted provides inspection outcomes and related statistics that can be aggregated by provider type or geography.

The theory's key objects (τ, t, x, y, and income heterogeneity) can be measured using a combination of ISC census data, DfE school census statistics, and ONS income estimates, with public-sector outcomes proxied using Ofsted inspection outcome data. These sources are appropriate because they align closely with the model's theoretical objects rather than relying on indirect proxies. While these measures are imperfect proxies for the model's theoretical objects, they capture the key variation relevant for evaluating the model's predictions.

3.3 [7.5 marks]

Consider Prediction 1: when the VAT-inclusive price of private schooling rises, private-school attendance falls.

Let r index local areas (e.g., local authorities) and t index years. Define:

$PrivateShare_{rt}$: share of pupils attending independent schools in the area r, year t (constructed from ISC and DfE enrolment counts). (Independent Schools Council, 2025; Department for Education, 2025).

$Post_t$: indicator equal to 1 after VAT is applied to private-school fees, zero before.

$Exposure_r$: pre-reform exposure to private schooling in the area r (e.g., baseline independent-school share).

X_{rt}: controls (e.g., local income from ONS, demographics).

δ_r: area fixed effects, λ_t: year fixed effects.

A feasible specification that exploits differential exposure is:

$$PrivateShare_{rt} = \alpha + \beta(Post_t \times Exposure_r) + \delta_r + \lambda_t + X'_{rt}\gamma + \varepsilon_{rt}.$$

This specification closely reflects the model's comparative statics since variation in τ It is policy-driven rather than chosen by schools or households. (Angrist and Pischke, 2009; Imbens and Wooldridge, 2009).

Interpretation of terms

$\delta_r$controls for time-invariant local differences in preferences or schooling culture (capturing persistent differences in P_i-type valuation across areas.

λ_t controls for national shocks and trends affecting all areas equally (macro conditions, national education reforms).

$Post_t \times Exposure_r$ captures that areas with higher baseline private-school reliance face a larger "effective shock" to the local private-school market after VAT.

Coefficient of interest and hypothesis. The coefficient of interest is β. The theory predicts $\beta < 0$: VAT increases the effective cost of private schooling and raises the relative attractiveness of public schooling, so high-exposure areas should experience larger declines in private-school share post-reform.

A key identifying assumption is that, absent the VAT reform, high- and low-exposure areas would have followed parallel trends in independent-school share. This can be assessed by plotting/event-studying pre-trends and estimating leads/lags of the interaction term.

Thus, the hypothesis test is:

$$H_0: \beta = 0 \text{ vs } H_A: \beta < 0.$$

This regression tests whether the VAT reform reduced private-school enrolment more in areas that were more exposed to private schooling before the reform, as predicted by the model's comparative statics. It mirrors the model's logic by exploiting exogenous variation in τ rather than endogenous variation in fees.

3.4 [7.5 marks]

Even with fixed effects, the regression may suffer from endogeneity for several reasons. This reflects the module's wider point that education outcomes are hard to interpret causally because private-school attendance is strongly selected on family characteristics and estimated 'private school effects' fall once income and parental education are controlled for.

i) Omitted variables correlated with exposure and trends

Areas with high baseline private-school shares may also differ in unobserved ways that evolve over time (e.g., local housing market dynamics, changing school quality, peer effects). In the theoretical model, these map into shifts in effective valuations P_i and in the attractiveness of public schooling, potentially making ε_{rt} correlated with $Exposure_r \times Post_t$.

ii) Simultaneity in fee-setting and demand. Although the Part 2 model treats t as a parameter, private schools may adjust fees in response to demand. High demand (unobserved parental preferences) can raise both fees and enrolment, biasing price-based estimates if one uses $t(1+\tau)$ directly as a regressor.

iii) Policy endogeneity and concurrent reforms

The timing of VAT could coincide with other national policy changes affecting schooling choices (public school funding reforms, bursary policies). If these reforms are correlated with $Post_t$, the estimated effect may conflate VAT with other changes.

Solutions and feasibility

Difference-in-differences with robustness checks

The exposure based DiD framework above is a practical way to mitigate simultaneity in fees because it relies on a policy discontinuity and pre-determined exposure rather than endogenous

contemporaneous fees. To support credibility, one would evaluate parallel pre-trends in private-school shares between high- and low-exposure areas.

Instrumental variables (IV) for price-based regressions

If the empirical approach uses the VAT-inclusive fee $t(1+\tau)$ directly, variation in τ induced by policy can serve as an instrument for the VAT-inclusive price component (conceptually consistent with the model where τ is policy-driven. The practicality depends on having sufficient variation and a strong first stage.

Augmenting controls and using outcomes data

Incorporating time-varying controls (income, demographics) and proxies for public-school quality (class size indicators, inspection outcomes) can reduce omitted variable bias.

The main threats are omitted variables, fee endogeneity, and contemporaneous reforms. A policy-based DiD strategy with exposure, supported by pre-trend checks and rich controls, is the most implementable approach in practice, with IV as a complementary strategy for price-based specifications. (Angrist and Pischke, 2009). These concerns highlight why policy-driven variation, rather than cross-sectional price difference, is the most credible way to assess the models' predictions. This reflects a broader theme discussed in the lectures, namely that policy evaluation must distinguish price effects from selection effects driven by underlying social and income heterogeneity.

Part 4 [15 marks]

The theoretical model developed in Part 2 provides a useful framework for evaluating several of the arguments raised in the two YouTube videos discussing the introduction of VAT on private school fees. In particular, the model clarifies the economic mechanisms linking higher private schooling costs, public school funding, and school choice behaviour.

In the course framing, the arguments in the videos can be organised around an equity–efficiency trade-off: who bears the burden of the tax and who benefits from the implied change in public funding, versus whether the policy creates distortions through switching and pressure on capacity.

A central claim made in the videos is that applying VAT to private school fees will increase the financial burden on families using private education and may push some pupils into the state sector. This mechanism is directly captured in the model through the VAT-inclusive price. $t(1+\tau)$, which lowers the utility of private schooling as the VAT rate increases. The model predicts that middle-income families are most likely to switch from private to public schooling, since they are closest to indifference between the two sectors. This aligns closely with concerns in the videos about families "at the margin" being disproportionately affected. The empirical strategy outlined in Part 3 allows this prediction to be assessed by examining whether private-school enrolment falls more sharply in areas with greater exposure to private education following the VAT reform. THIS shows a contrast between theoretical mechanisms and real-world complexity reflects the course's broader emphasis, which is a good example of the course's broader emphasis on linking a positive mechanism (how incentives change) to an empirical strategy (what we would observe in data), before moving to normative claims about whether the policy is 'good'.

The videos also argue that VAT revenue could improve state school funding but may be insufficient to offset the pressure created by increased demand. This trade-off is reflected in the public-school payoff term. $S/x + 5\tau ty/x$, which shows that funding gains from VAT are shared among public-school pupils and therefore depend on enrolment. However, the model abstracts from short-run capacity constraints, teacher shortages, and adjustment costs, all of which are emphasised in the videos but not explicitly modelled. (Sibieta, 2025). This matters because, as discussed in the education lecture, state-school running costs are funded from general taxation (government budget), so the policy effectively shifts part of education financing from general revenues towards a tax concentrated on private-school users. This mirrors broader debates on state capacity, where the effectiveness of the redistributive policy depends not only on revenue raised but also on institutional ability to absorb increased demands.

Several arguments raised in the videos, therefore, fall outside the scope of the current framework. These include transitional disruption effects, heterogeneity in public-school quality, and broader social considerations such as peer effects. (Hoxby, 2000). To analyse these issues, the model could be extended by allowing public-school quality to depend non-linearly on enrolment, or by introducing capacity constraints in the public sector. Another extension would be to introduce dynamic adjustment costs, distinguishing short-run disruptions from long-run equilibria. (Hoxby, 2000).

What I find most persuasive in the videos is the focus on the 'squeezed middle' response: the model makes it clear that the largest behavioural movement should come from families close to indifference, not from the very highest-income households. Overall, the model provides a disciplined way to interpret the main economic arguments raised in the videos, while also highlighting which elasticity of concerns require additional modelling beyond the current framework.

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